

Ex 1 Study the nature of the following integrals.

③ 1. $\int_0^{+\infty} \frac{\arctan x}{\sqrt{x}} dx$; mean 0⁺: $\frac{\arctan x}{\sqrt{x}} \sim \frac{\pi}{\sqrt{x}} = \frac{1}{x^{1/2}} > 0$ Convergent (By Riemann & equiv. test).
 mean ∞ : $\frac{\arctan x}{\sqrt{x}} \sim \frac{\pi/2}{\sqrt{x}} > 0$ Convergent (Riemann & equiv. test).

Hence the imp. integ 1. is conv.

③ 2. $\int_1^{+\infty} \frac{\cos(\ln x)}{x} dx$; $t = \ln x = u(x)$ of class C^1 on $[1, +\infty[$
 $g(t) = \cos t$ is cont. on $u([1, +\infty[) =]0, +\infty[$.
 $\stackrel{(1)}{=} \int_0^{+\infty} \cos t \, dt = \sin t \Big|_0^{+\infty} \stackrel{(2)}{=} \dots$ Hence the imp. int 2. is div.

③ 3. $\int_0^{+\infty} x \sin(\frac{1}{x}) dx$; mean 0⁺: $x^{1/2} [x \sin \frac{1}{x}] \xrightarrow{x \rightarrow 0^+} 0$, $\alpha = 1/2 < 1$ (absol conv).
 mean ∞ : $x \sin \frac{1}{x} \sim \frac{x}{1+x^2} = \frac{1}{x^2} > 0$ Conv (Riemann & equiv. test).

Thus 3. is conv.

③ 4. $\int_1^{+\infty} \frac{1}{\sqrt{x-1} \sqrt{x}} dx$; mean 1⁺: $\frac{1}{\sqrt{x-1} \sqrt{x}} \sim \frac{1}{(u-1)^{1/2}}$ conv (Riemann)
 mean 2⁺: $\frac{1}{\sqrt{x-1} \sqrt{x}} \sim \frac{1}{(2-x)^{1/2}}$ conv (Riemann)

Thus 4. is conv.

③ 5. $\int_0^1 \frac{x}{\ln x} dx$; mean 0⁺: $\frac{x}{\ln x} = \frac{1}{u^{1/2} \ln u}$ conv (Riemann)
 mean 1⁺: $\frac{x}{\ln x} = \frac{2}{\ln(1+(u-1))} \sim \frac{2}{1-(u-1)^2}$ div (Riemann)

Thus 5. div.

③ 6. $\int_{-\pi/2}^{\pi/2} \ln(1+\sin x) dx$ has same nature as $\int_{-\pi/2}^0 \ln(1+\sin x) dx$

$\int_{-\pi/2}^0 \ln(1+\sin x) dx = \int_{-\pi/2}^0 \frac{\tilde{g}(u(x)) \cdot \tilde{g}'(x)}{g(x)} dx$; $t = 1+\sin x = u(x)$ of class C^1 on $]-\pi/2, 0[$
 $g(t) = \frac{\ln t}{\sqrt{t(t-1)}}$ and on $u(]-\pi/2, 0[) =]0, 1[$
 $\stackrel{(1)}{=} \int_0^1 \frac{\ln t}{\sqrt{t(t-1)}} dt$; mean 0⁺: $\frac{\ln t}{\sqrt{t(t-1)}} \sim \frac{\ln t}{\sqrt{t}} = \frac{1}{t^{1/2}} \cdot \frac{1}{t^{1/2}} < 0$ conv (p. 1)

Study the nature of the following series.

① $\sum_{n=1}^{\infty} (1 + \frac{1}{n^n})$; $(1 + \frac{1}{n^n}) \xrightarrow{n \rightarrow \infty} 1 \neq 0$, then the series div.

② $\sum_{n=1}^{\infty} (2^n - 1)^n$; $[(2^n - 1)^n]^{\frac{1}{n}} = 2^n - 1 = e^{\frac{\ln n}{n}} - 1 \xrightarrow{n \rightarrow \infty} 0 < 1$ (Conv)

③ $\sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{n}$; $\frac{\sqrt{n+1} - \sqrt{n}}{n} \times \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{n(\sqrt{n+1} + \sqrt{n})} \sim \frac{1}{n[2\sqrt{n}]} = \frac{1}{2n^{3/2}} \xrightarrow{n \rightarrow \infty} 0$ (Conv)
 equivalent to $\frac{1}{n^{3/2}}$

④ $\sum_{n=2}^{\infty} \frac{(-1)^n}{n(\ln n)^2}$; $\left\{ \frac{1}{n(\ln n)^2} \right\}_{n=2}^{\infty}$ and $\xrightarrow{n \rightarrow \infty} 0$ (Conv) Alternating series test

⑤ $\sum_{n=1}^{\infty} \frac{\cos \sqrt{n} \sin n^2}{n\sqrt{n}}$; $0 \leq \left| \frac{\cos \sqrt{n} \sin n^2}{n\sqrt{n}} \right| < \frac{1}{n^{3/2}}$ (absol conv) Dirichlet's test

⑥ $\sum_{n=1}^{\infty} \ln(\cos \frac{1}{n}) \ln(\sin \frac{1}{n})$; $\ln(\cos \frac{1}{n}) \ln(\sin \frac{1}{n}) \sim \ln(1 - \frac{1}{n^2}) \ln(\frac{1}{n}) \sim \frac{(-1/n^2)(-\ln n)}{n^2} = \frac{\ln n}{n^2} > 0$
 conv (Leibniz)

Ex 1. let $S_n = \sum_{k=1}^n \frac{1}{n+k}$ and $T_n = \sum_{k=1}^n \frac{(-1)^{k-1}}{k}$

① Calculate $\lim_{n \rightarrow \infty} S_n$.

$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n+k} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(\frac{k}{n})$; $f(x) = \frac{1}{1+x} \in C([0,1]) \subseteq R([0,1])$
 $= \int_0^1 f(x) dx = \int_0^1 \frac{1}{1+x} dx = \ln(1+x) \Big|_0^1 = \ln 2$.

② S.T. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ is semi-convergent.

- The series converges by Alternating series test ($\frac{1}{n}$ pos, dec $\xrightarrow{n \rightarrow \infty} 0$).
- The series is not absol conv by Leibniz's test.

③ S.T. $T_{2n} = \sum_{k=n+1}^{2n} \frac{1}{k}$.

$T_{2n} = \sum_{k=1}^{2n} \frac{(-1)^{k-1}}{k} = \sum_{k=1}^{2n} \frac{1}{k} - 2 \sum_{k=1}^n \frac{1}{2k} = \sum_{k=n+1}^{2n} \frac{1}{k}$

Deduce that $T_{2n} = S_n$.

$T_{2n} = \sum_{k=n+1}^{2n} \frac{1}{k} = \sum_{k=n+1-n}^{2n-n} \frac{1}{n+k} = \sum_{k=1}^n \frac{1}{n+k} = S_n$ (By shifting index)

④ Deduce the value of the sum $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$.

$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \stackrel{①}{=} \lim_{n \rightarrow \infty} T_n \stackrel{②}{=} \lim_{n \rightarrow \infty} T_{2n}$ (since T_n conv by 2, then any subseq. conv to same limit)
 $\stackrel{③}{=} \lim_{n \rightarrow \infty} S_n \stackrel{④}{=} \ln 2$.

Assume; $f: [0,1] \rightarrow \mathbb{R}$ continuous
 $g: \mathbb{R} \rightarrow \mathbb{R}$ cont, $g(x) \geq 0$ and $g(x+1) = g(x) \forall x \in \mathbb{R}$.

① 1. S.T $\int_0^1 f(x)g(nx)dx = \frac{1}{n} \sum_{k=1}^n \int_0^1 f(\frac{k-1+s}{n})g(s)ds$

$\frac{1}{n} \sum_{k=1}^n \int_0^1 f(\frac{k-1+s}{n})g(s)ds$; $x = \frac{k-1+s}{n}$, $s = nx - k + 1$ & $ds = ndx$

① $\frac{1}{n} \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} f(x)g(nx-k+1)ndx = \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} f(x)g(nx)dx$

① $\int_0^{\frac{1}{n}} f(x)g(nx)dx + \int_{\frac{1}{n}}^{\frac{2}{n}} f(x)g(nx)dx + \dots + \int_{\frac{n-1}{n}}^1 f(x)g(nx)dx = \int_0^1 f(x)g(nx)dx$

② 2. S.T $\exists s_k \in [0,1]$ s.t $\int_0^1 f(x)g(nx)dx = \frac{1}{n} \sum_{k=1}^n f(\frac{k-1+s_k}{n}) \int_0^1 g(s)ds$

$\int_0^1 f(x)g(nx)dx = \frac{1}{n} \sum_{k=1}^n \int_0^1 h(s)g(s)ds$; $h(s) = f(\frac{k-1+s}{n}) = f(u(s))$
 $u(s) = \frac{k-1+s}{n}$

$= \frac{1}{n} \sum_{k=1}^n \int_0^1 g(s)ds$ By 1st M.V.T:
 $\bullet f$ cont on $[0,1]$ (composed of cont f & u)
 $\bullet g$ has const sign on $[0,1]$

① $= \frac{1}{n} \sum_{k=1}^n f(\frac{k-1+s_k}{n}) \int_0^1 g(s)ds$

③ 3. S.T $\lim_{n \rightarrow \infty} \int_0^1 f(x)g(nx)dx = \int_0^1 f(x)dx \int_0^1 g(x)dx$

$\lim_{n \rightarrow \infty} \int_0^1 f(x)g(nx)dx = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(\frac{k-1+s_k}{n}) \int_0^1 g(s)ds$

$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(c_k) \int_0^1 g(s)ds$

If we choose the regular partition $P_n = \{x_k = \frac{k}{n}; 0 \leq k \leq n\}$
 $c_k = \frac{k-1+s_k}{n} \in [x_{k-1}, x_k]$ ($\because 0 \leq s_k \leq 1 \Rightarrow \frac{k-1}{n} \leq \frac{k-1+s_k}{n} \leq \frac{k-1+1}{n} = \frac{k}{n}$
 $\Rightarrow x_{k-1} \leq c_k \leq x_k$)

① $\int_0^1 f(x)dx \int_0^1 g(x)dx$ ($\bullet f \in C([0,1]) \subseteq R([0,1])$
 $\bullet c_k \in [x_{k-1}, x_k]$)

4. Assume $g(x) \in \mathbb{R}$ & $c = \min_{x \in \mathbb{R}} g(x)$ & $h(x) = g(x) - c$

② (a) $c = \min_{x \in \mathbb{R}} g(x) \Rightarrow c \leq g(x) \forall x \in \mathbb{R} \Rightarrow h(x) = g(x) - c \geq 0 \forall x \in \mathbb{R}$

③ (b) $\lim_{n \rightarrow \infty} \int_0^1 f(x)h(nx)dx = \int_0^1 f(x)dx \int_0^1 h(x)dx$ bcz $\bullet h(x) \geq 0$ & $h(x+1) = h(x) \forall x \in \mathbb{R}$
 $\Rightarrow \lim_{n \rightarrow \infty} \int_0^1 f(x)g(nx)dx = c \int_0^1 f(x)dx + \lim_{n \rightarrow \infty} \int_0^1 f(x)h(nx)dx = c \int_0^1 f(x)dx + \int_0^1 f(x)dx \int_0^1 h(x)dx$

③ 5. Define $\lim_{n \rightarrow \infty} \int_0^1 f(x) \sin(2n\pi x)dx = \int_0^1 f(x)dx \int_0^1 \sin(2\pi x)dx$; bcz $g(x) = \sin(2\pi x)$
 $\bullet g$ cont